Illuminating the Unknown

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University of Sydney - Honours Thesis

THE UNIVERSITY OF SYDNEY

MOTIVATION: NAVIGATING IN THE DARK

Challenge vs Opportunity

Challenge	Opportunity
Reduce feature contrast and descriptor repeatability undermine tracking robustness.	Enable UAV navigation in low-light, GPS-denied, indoor environments.
SLAM becomes unstable in texture-poor or dark environments, causing drift or tracking loss.	Integrate Low-Light Image Enhancement (LLIE) to restore perceptual / geometric detail for stable SLAM.

Purely visual SLAM systems suffer severe degradation in low-light, GPS-denied environments, yet they remain the most lightweight and cost-effective option for UAVs.

RESEARCH GOAL

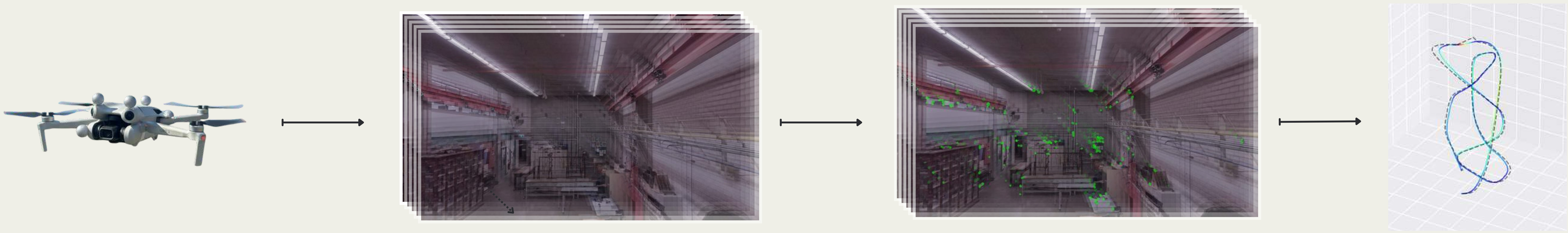
Aim

- To experimentally evaluate real time UAV SLAM performance in low-light environments.

Objectives

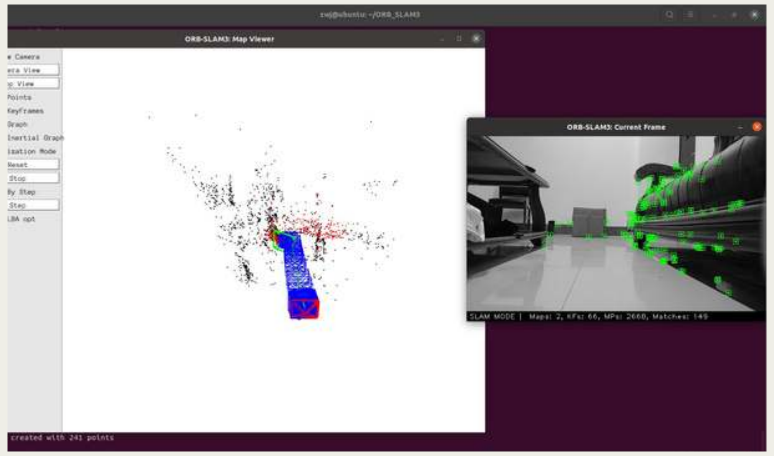
- Assess how reduced illumination affects feature extraction, tracking, and localisation accuracy.
- Integrate and test real-time Low-Light Image Enhancement (LLIE) models within the SLAM pipeline.
- Quantify the impact on trajectory accuracy and tracking robustness under controlled conditions.
- Comprehensively review existing methods and suggest future direction for improved low light SLAM.

WHAT IS SLAM?



Simultaneous Localisation and Mapping - estimating both the camera's position and the map of the environment in real-time.

LITERATURE REVIEW



SLAM Fundamentals

Why ORBSLAM3

Provides a controlled benchmark to isolate how illumination directly impacts tracking and mapping.

Low Light Limitations

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Reduced contrast and rising noise drastically lower keypoint detection and inlier ratios.
Without stable features, monocular SLAM loses scale, drifts, or resets entirely.

Integration Approaches

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Two paradigms dominate. sequential enhancement before SLAM, or tightly-coupled joint optimisation.
Studies have found that while LLIE improves visual appeal, this does not translate to improved trajectory accuracy.
But why not?

Research Gap

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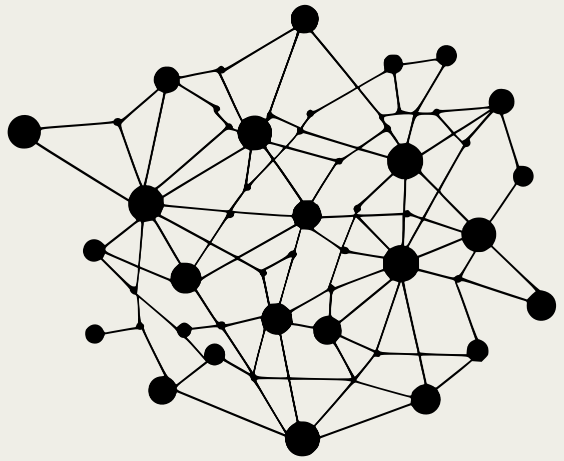
Existing work focuses on perceptual metrics, not SLAM's internal performance.
This thesis evaluates LLIE's real effect on feature quality, inliers, and tracking stability within ORB-SLAM3.

L I T E R A T U R E   R E V I E W

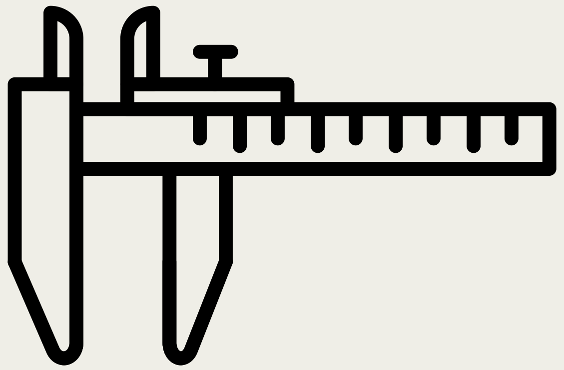
From Classical Illumination to Neural Enhancement - the Foundations of Modern LLIE



Learning Based

Zero-DCE++

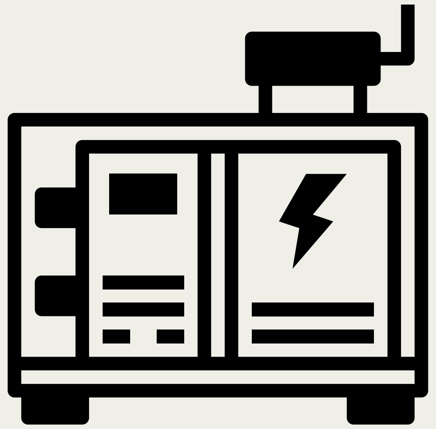
A lightweight deep curve estimation network that directly adjusts pixel illumination without paired training data. Known for real-time performance.



Self Calibrated

SCI

Introduces a self-supervised structure that selectively illuminates details that are typically lost in low light frames.



Generative

EnlightenGAN

Leverages adversarial training to map dark images to bright domains while preserving texture detail. Computationally heavier for real-time use.

Classical Baselines

CLAHE/HE

Non-learning enhancement that redistributes brightness histograms to increase contrast. Provides a benchmark for evaluating the benefit of deep LLIE models.

EXPERIMENTAL FRAMEWORK

Adapt
Understand the Problem

Reviewed degradation of ORB-SLAM3 under low-light and identified feature loss as the dominant limitation.

Evaluate
Model Benchmarking

Compared LLIE algorithms on public datasets using Trajectory and fundamental Computer Vision metrics to establish quantitative baselines.

Launch
Pipeline Integration

Integrated LLIE modules into a real-time ROS pipeline feeding ORB-SLAM3. Conducted Bennett Lab experiments with ROS Gazebo used for pipeline development.

Monitor
Performance Analysis

Measured ORB feature counts, tracking ratios, and ATE/RPE trajectories to assess how enhancement influences localisation accuracy.

Iterate
Refinement and Expansion

Summarise findings to motivate and assist future direction.
Stereo and depth sensors.
Temporally aware enhancement.

PROCESS

Test Pipeline Feasibility

Low light models developed for single frame processing.

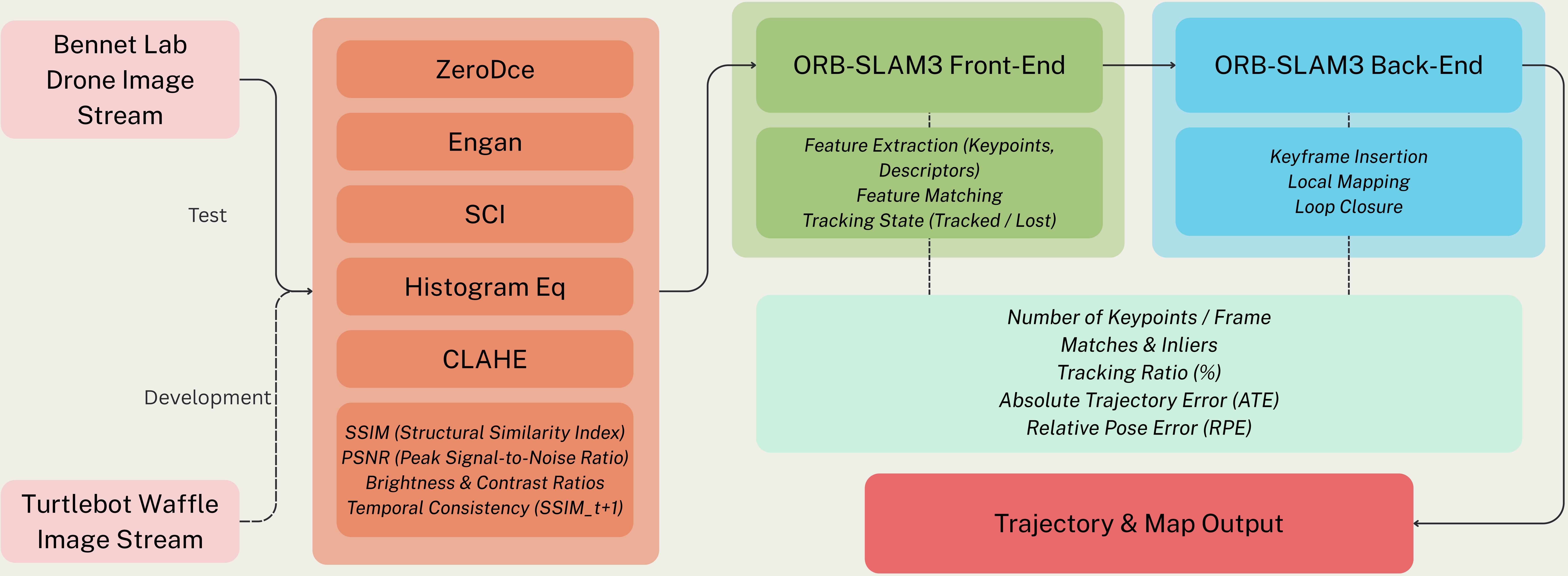
Record Dataset

Lack of publicly available UAV low light datasets with groundtruth data.
Contribute to the robotics community

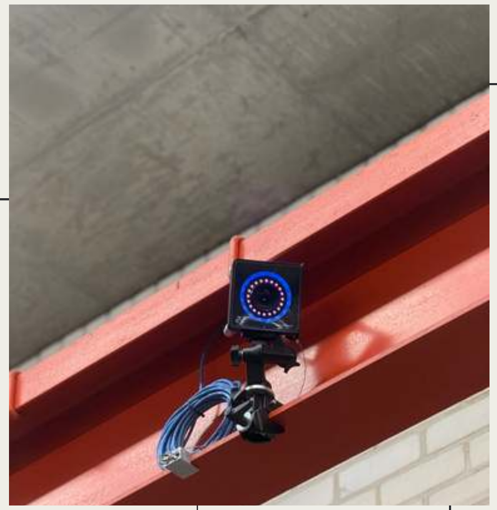
Evaluate

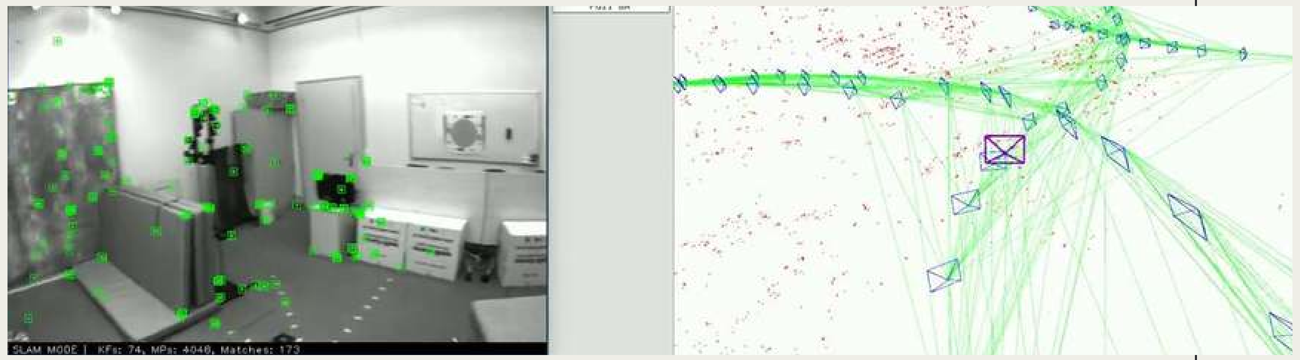
Combine pipeline with dataset.
Evaluate SLAM metrics etc.

PIPELINE


Bennet Lab Drone Image Stream
Test
Development
Turtlebot Waffle Image Stream
ZeroDce
Engan
SCI
Histogram Eq
CLAHE
SSIM (Structural Similarity Index)
PSNR (Peak Signal-to-Noise Ratio)
Brightness & Contrast Ratios
Temporal Consistency (SSIM_t+1)
ORB-SLAM3 Front-End
Feature Extraction (Keypoints, Descriptors)
Feature Matching
Tracking State (Tracked / Lost)
ORB-SLAM3 Back-End
Keyframe Insertion
Local Mapping
Loop Closure
Number of Keypoints / Frame
Matches & Inliers
Tracking Ratio (%)
Absolute Trajectory Error (ATE)
Relative Pose Error (RPE)
Trajectory & Map Output

EXPERIMENTAL SETUP





Datasets

Bennet Lab

Fundamental source of truth.
Primary dataset collection point.

UAV

DJI Mini 4 Pro

Lightweight, safe and easy to operate drone
High resolution, modifiable, realistic.

Tracking

OptiTrack Camera

26 cameras with millimetre precision ground truth position and rotation tracking

Synthesis

ORBSLAM

Chosen as the SLAM backbone.

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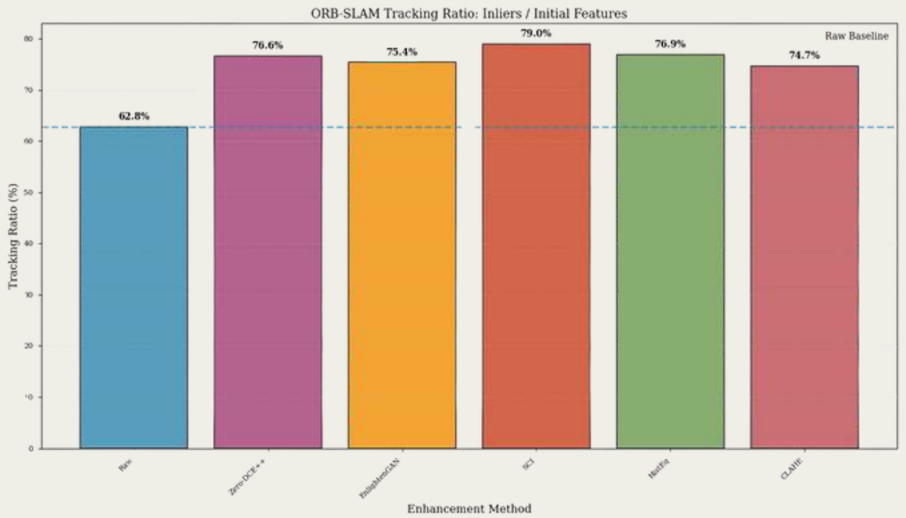
SCI

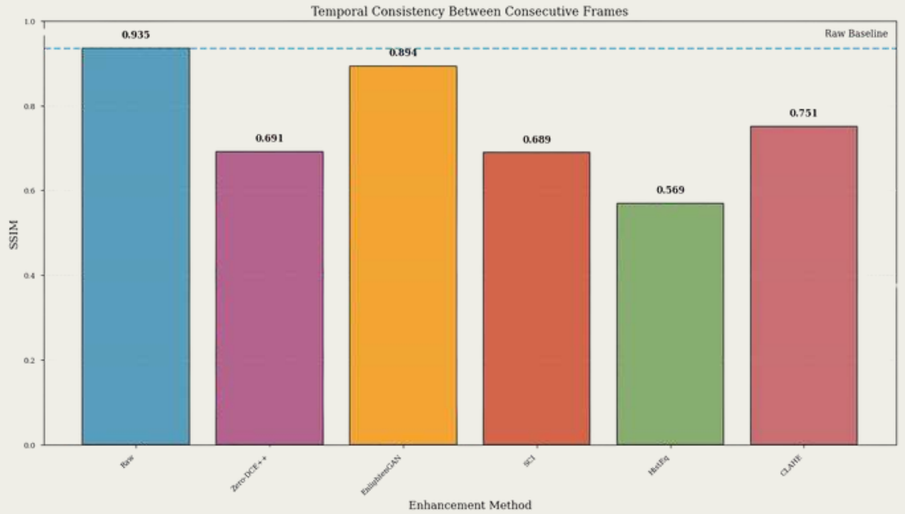
EnlightenGAN

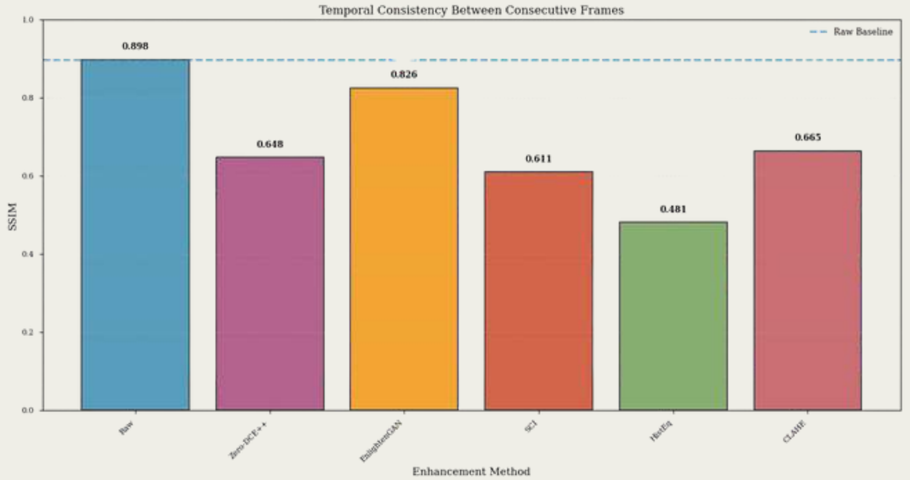
CLAHE

ZeroDCE++

Histogram Equalisation





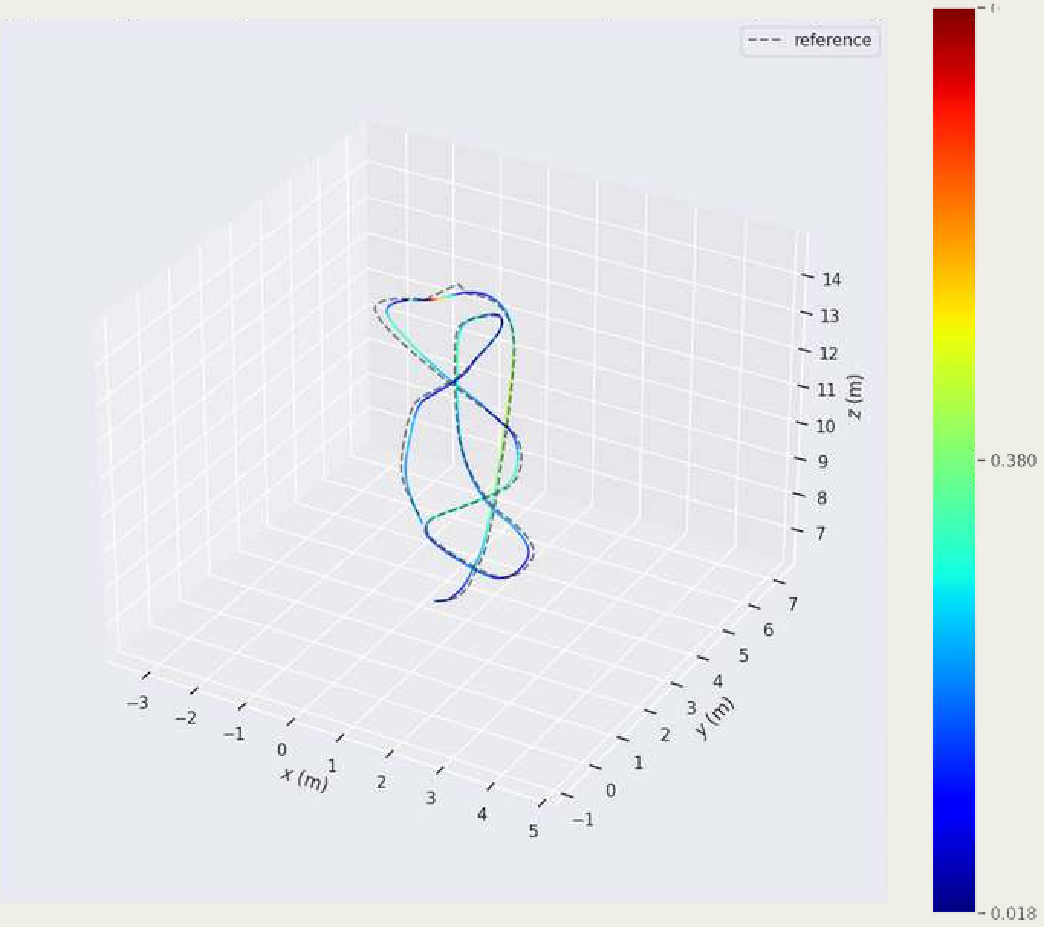


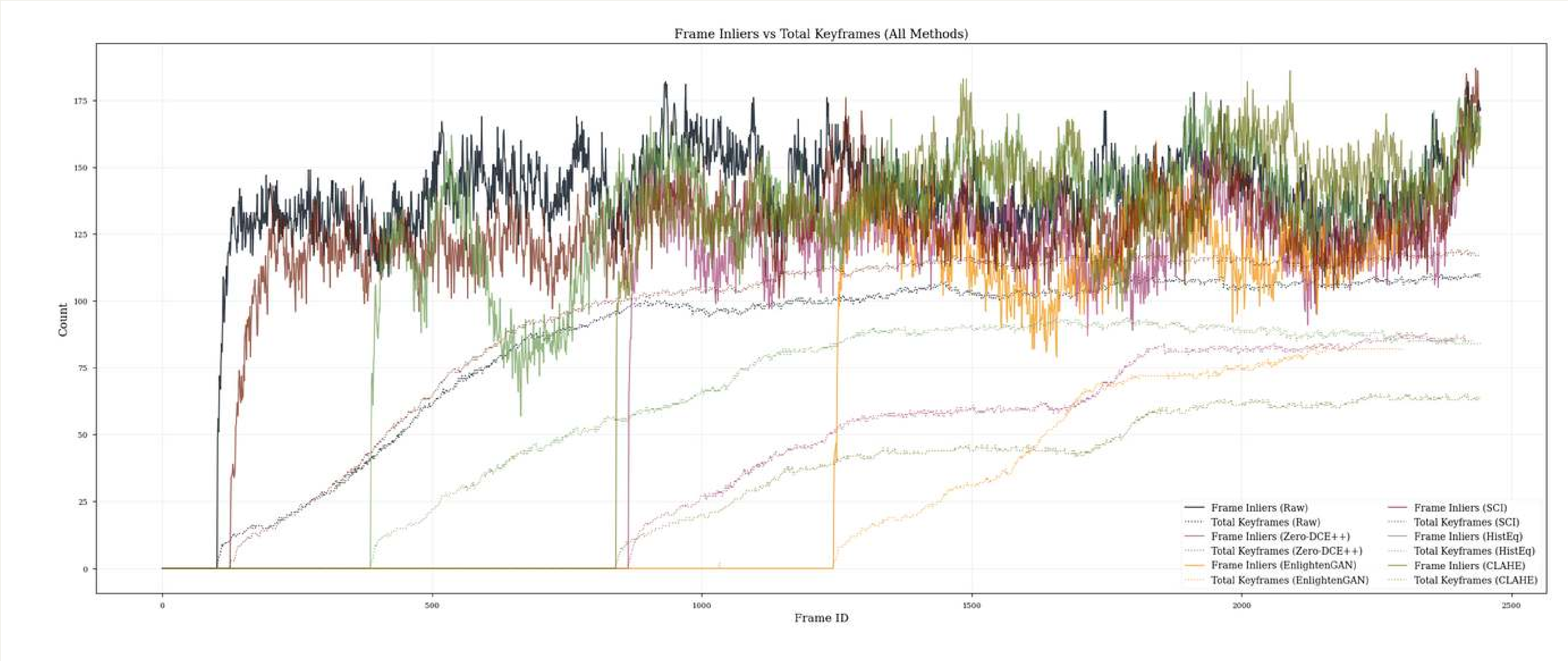
raw,sci, dce, engan, clahe, he

RESULTS - TRAJECTORY

Mean APE Comparison (m)

Method	Take 10	Take 11	Take X
Raw	tbd	**0.233**	
ZeroDce	tbd	1.993	
EnGan	tbd	2.52	
SCI	tbd	0.236	
HE	tbd	1.899	
CLAHE	tbd	2.09	





Take 11
1/200s Shutter
Lights off
'Raw Low Light'

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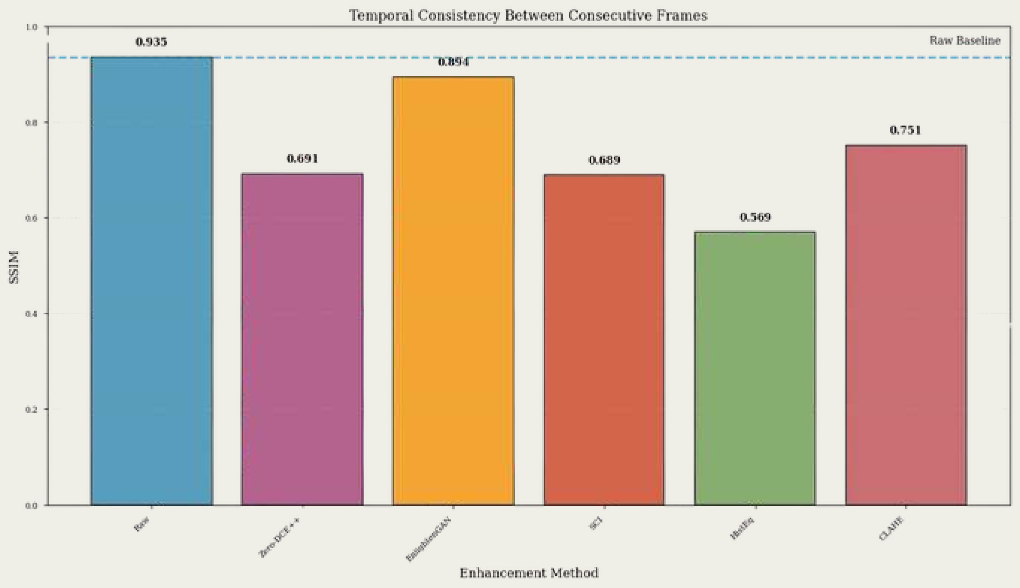
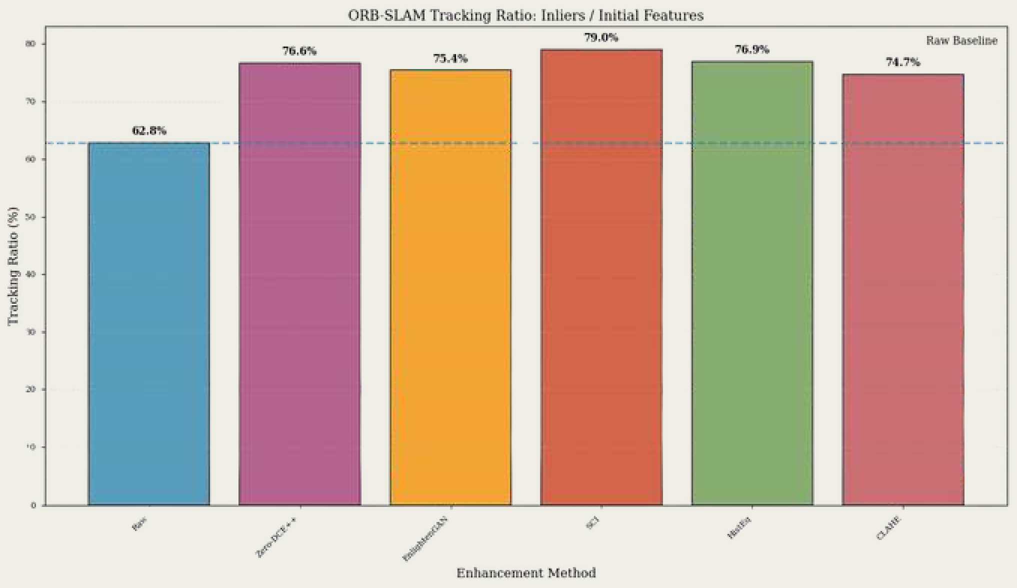
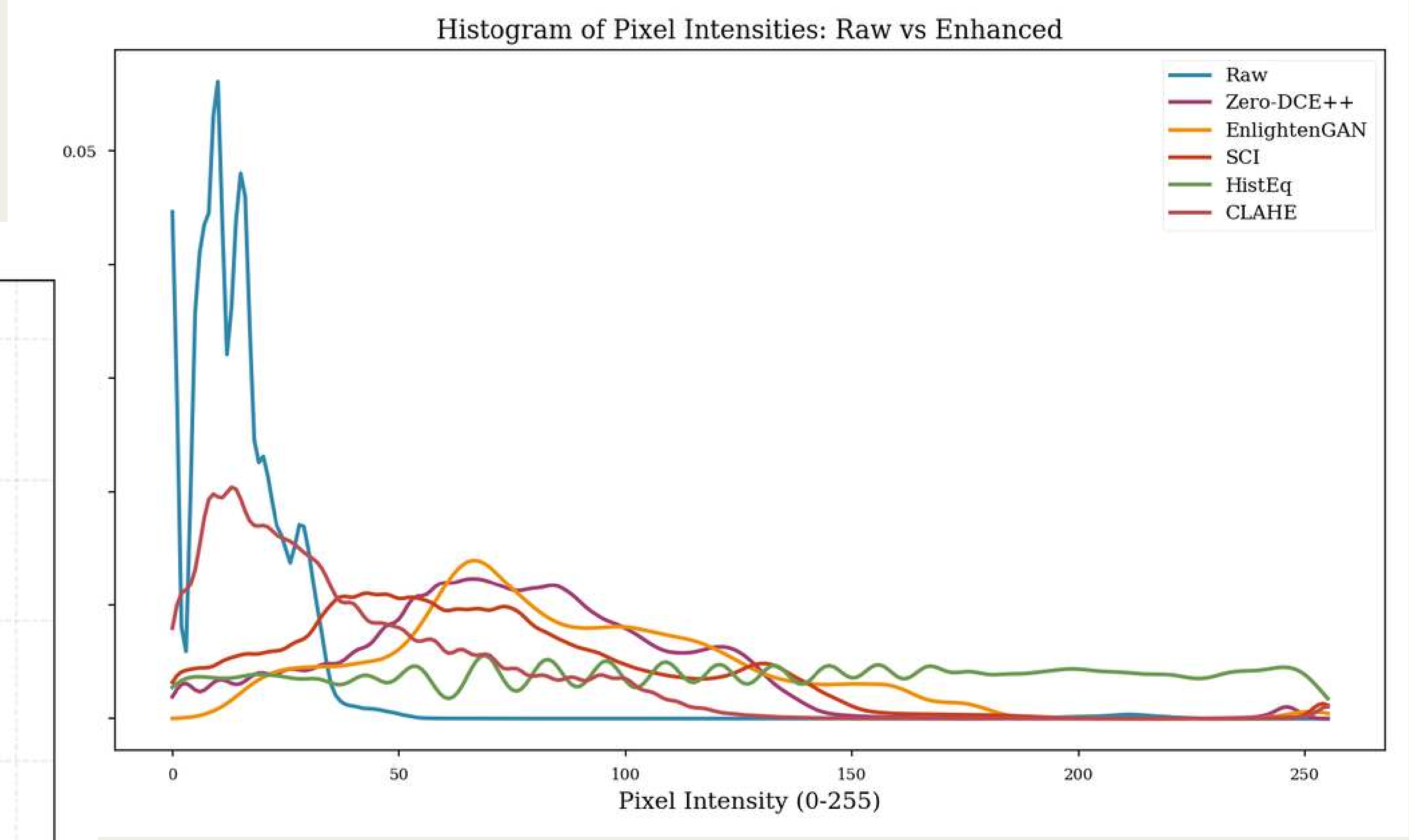
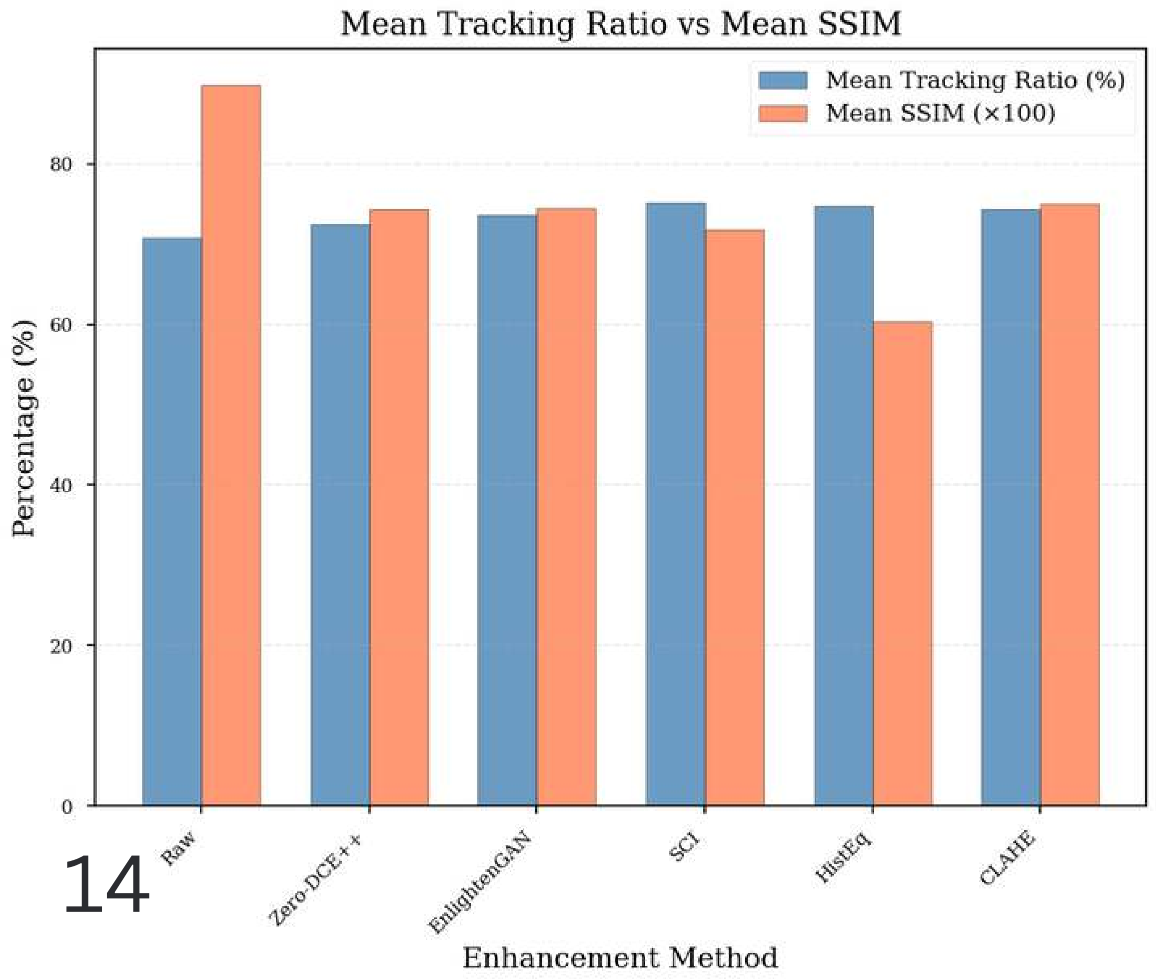
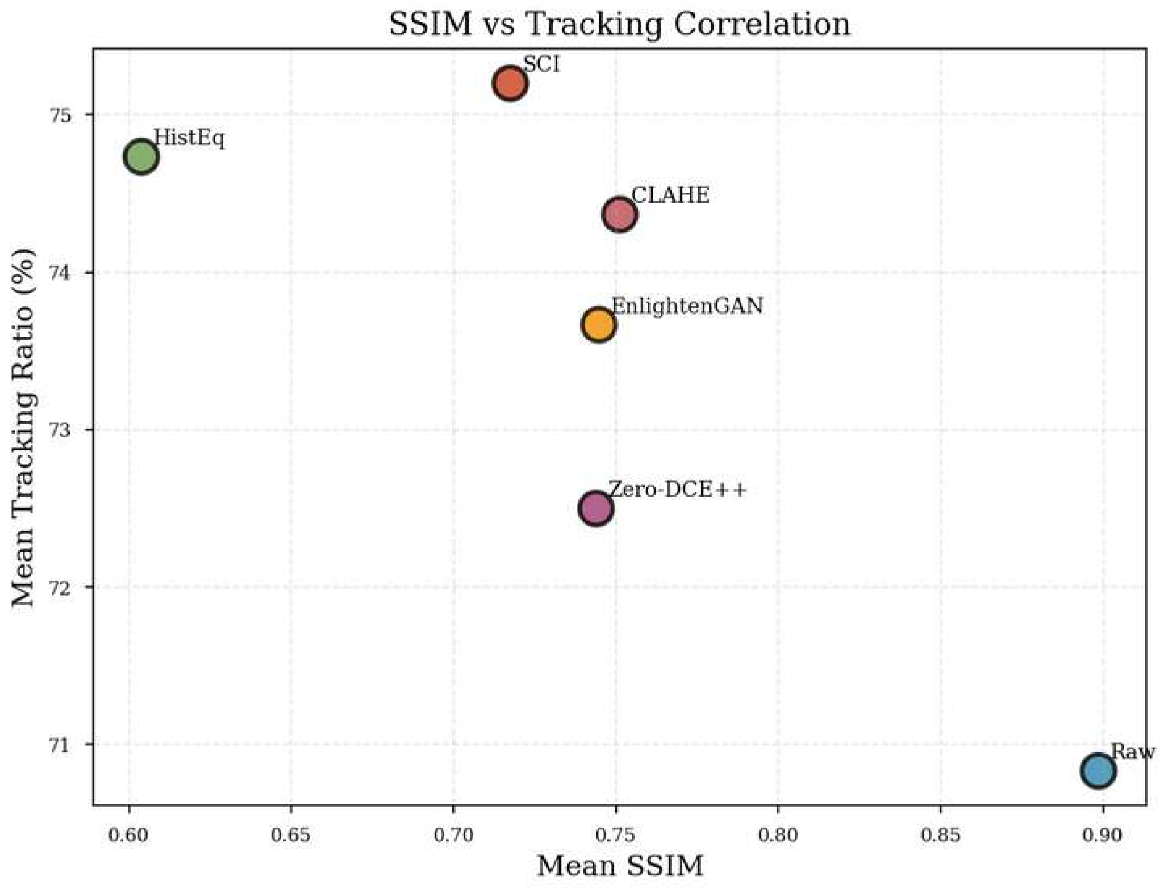
RESULTS











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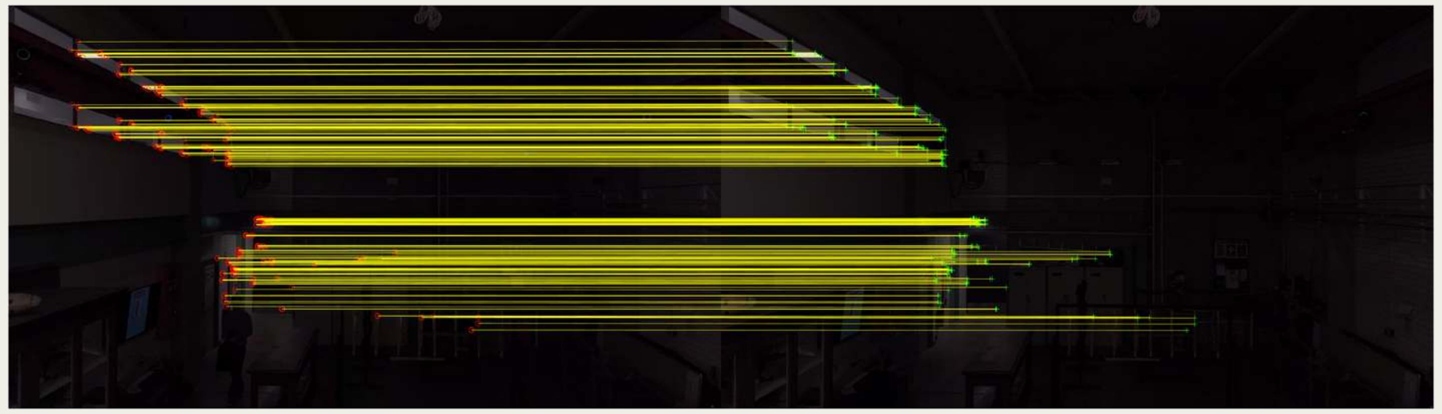
RESULTS - VISION



DISCUSSION

lipsum

FUTURE WORK

Varying SLAM Architectures with Fixed LLIE

Future research could reverse the current focus by fixing the enhancement model and testing various SLAM backbones like DSO, SuperPoint, or visual-inertial odometry. This approach would help assess how different SLAM architectures perform feature extraction on enhanced imagery and determine their resilience to photometric distortions caused by LLIE.

Temporal Stability Analysis

Current LLIE evaluation was performed on a frame-by-frame, realtime basis. Due to real-time SLAM sensitivity to temporal consistency between consecutive frames, future analysis should investigate methods to introduce robust temporal stability.

Integration of Additional Sensors (Stereo/Depth)

Expanding beyond monocular input, future work should incorporate stereo vision and depth sensors (e.g., Intel RealSense D435i). Combining LLIE with multi-view geometry could improve depth estimation and feature triangulation in low-light conditions, providing richer scene structure and reducing scale drift in dark or texture-poor environments.

Real-World Deployment Dataset Expansion

Dataset Expansion

Semantic Integration and Detection

CONCLUSION

Built a Real-World Low-Light Dataset

Captured a controlled UAV dataset in the Bennett Lab, providing new low-light sequences to expand open-source resources for future SLAM and LLIE research.

Fused LLIE with Visual SLAM

Integrated state-of-the-art low-light image enhancement models into a live ORB-SLAM3 ROS pipeline - bridging two traditionally separate areas of computer vision.

Implemented a fully modular ROS-based framework capable of swapping enhancement modules and SLAM components in real-time for comparative evaluation.

Quantitatively Benchmarked Enhancement Impact

Showed that while LLIE improves perceived visibility, ORB-SLAM3 remains remarkably resilient in low light, highlighting its photo-metric stability.

Revealing thresholds where it begins to fail.

Questions?

Thank you.

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